LESSON 11 · MISSION GUIDE

Before the Throne

Charts people can read, and Excel files people can open.

From **Queen Aldara of Thornmere**
To **Jen, Keeper of the Ledger**
By raven **at dusk, on the last day of the week**

Keeper, Mistress Pell has shown us what you have been doing with the "pandas". Most impressive. We understood almost half of it.

The Privy Council sits in a fortnight. The Privy Council does not read tables. It reads *pictures*, ideally blue ones, and then asks for "the workbook" to fiddle with in the carriage home. Dalvric once drew the pictures by hand, by candlelight, for a whole moon, which may be why he sailed west. Can your Python draw them and make the workbook? With bold headings? No pressure. (Some pressure.) — A.

What you'll be able to do after today

- [ ] Draw a **bar chart** and a **histogram** from a DataFrame with matplotlib, and save them as PNG files
- [ ] Put two series **side by side** (men vs women per guild) and label everything properly
- [ ] Write a DataFrame to an **Excel workbook**, and several tables to several **sheets** of one workbook
- [ ] Format a workbook with openpyxl: bold coloured headers, `#,##0.00 "cr"` number formats, column widths, frozen panes
- [ ] Drop a chart image into a worksheet

Why this matters at work

- **The last mile is the one people see.** Nobody on the Privy Council will run your script. They'll open a PNG and an xlsx in the carriage home. If that looks right, the whole project looks right.
- **Formatting by code is formatting forever.** Set the header style once in Python and every month's report has it. No more "can you make it look like last month's".

Four big ideas

1. A chart is a file	In a script there is no window. `plt.savefig("x.png")` writes the picture; you open it like any file. `matplotlib.use("Agg")` at the top tells matplotlib not to look for a screen.
2. Build, then decorate	Order of a chart: `plt.figure()` → draw (`bar`, `hist`) → title, labels, legend → `tight_layout()` → `savefig()`. Same every time.
3. pandas writes Excel	`df.to_excel(path, sheet_name=...)` for one sheet; `with pd.ExcelWriter(path) as writer:` for several. Needs openpyxl installed, which is why it's in `requirements.txt`.
4. openpyxl edits the workbook	pandas writes values. For looks, re-open with `load_workbook`, walk the cells, set `font`, `fill`, `number_format`, widths, then `save`.

Things that will bite you

- Calling `plt.show()` in a script: it either hangs or does nothing. Save instead.
- Drawing a second chart without `plt.figure()`: both charts land on the same picture.
- Excel number formats are text: `'#,##0.00 "cr"'`. Exactly the string you'd type in Format Cells (the inner quotes are part of it).
- openpyxl colours have no `#`: `"1F3A5F"`. matplotlib colours do: `"#1f3a5f"`. Yes, really.

How today works

15 MIN
Lecture
Lars draws two charts and writes one workbook, live

60 MIN
Exercise sheet
Seven ravens: three charts, four workbooks

15 MIN
Pages from the Grimoire, page 11
The Council pack in one click. The Queen clicked.

The ravens

#	FROM	RAVEN	YOU'LL USE
1	Queen Aldara	Your first chart	`plt.bar savefig`
2	Lady Siv	Where do salaries sit?	`plt.hist axvline legend`
3	Mistress Pell	Side by side	`pivot_table np.arange` grouped bars
4	Orrin	Back to Excel	`to_excel`
5	Mistress Pell	One workbook, several sheets	`pd.ExcelWriter`
6	Queen Aldara	Make it pretty	`load_workbook Font PatternFill number_format`
7 ⭐	Queen Aldara	A chart inside the workbook	`barh add_image`

You're done when...

- [ ] `my_work/output/` holds four PNGs and four workbooks, and you've opened every one of them
- [ ] Every script prints what the sheet says, and Dalvric's Council pack names the right top three
- [ ] Everything is pushed to `lesson-11`, with your favourite chart attached to the pull request



Next lesson: The Grimoire Closes. The finale. Two sessions, one report: the Accord of Equal Coin, by the book, with a chart, a workbook, and one sentence the Queen can say to the Privy Council. Everything you've learned, in one project folder.